

SOC ANALYST

Complete Learning & Tools Handbook

A practical guide to SOC operations, SIEM, EDR, networking, Windows/Linux security, cloud, threat intelligence, detection engineering, incident response, automation, labs, commands, and interviews.

Scope: Beginner → Job-ready SOC Analyst → Advanced Blue Team foundations

Note: This handbook covers the major SOC tool categories and widely used examples; no single document can literally contain every commercial security product.

1. SOC Analyst at a Glance

- Mission: detect, investigate, contain, document, and help prevent security incidents.
- Workflow: telemetry → alert → triage → enrichment → investigation → containment/escalation → remediation → lessons learned.
- Typical environments: endpoints, identity, network, email, SaaS, cloud, applications, and security appliances.
- Typical levels: L1 / Junior SOC Analyst, L2 / Incident Responder, L3 / Detection Engineer or Threat Hunter.
- The transferable skill is understanding security data and investigative reasoning—not memorizing every product.

Core competency map

Area	What you should know	Priority
Networking	TCP/IP, DNS, HTTP/S, TLS, ports, routing, VPN, firewalls	Critical
Windows	Event logs, PowerShell, processes, services, registry, AD basics	Critical
Linux	auth logs, processes, permissions, systemd, SSH, shell	High
SIEM	search, parsing, correlation, dashboards, alert triage	Critical
EDR/XDR	process trees, persistence, isolation, detections	Critical
Threat Intel	IOCs, TTPs, enrichment, ATT&CK;	High
IR	containment, evidence, timelines, recovery, reporting	Critical
Cloud	IAM, audit logs, network controls, cloud detection basics	High
Automation	Python/PowerShell/Bash, APIs, SOAR concepts	Medium/High
Communication	tickets, incident notes, escalation, executive summaries	Critical

2. SOC Architecture & Roles

Typical SOC architecture

Log sources → collectors/agents → normalization → SIEM/data lake → detection rules → alert queue → analyst investigation → case management/SOAR → response systems.

Telemetry commonly comes from Windows Event Logs, Sysmon, Linux logs, firewalls, VPNs, proxies, DNS, DHCP, IDS/IPS, EDR, email security, identity providers, cloud audit logs, SaaS applications, and application servers.

Roles

L1: alert monitoring, triage, enrichment, basic investigation, ticketing, escalation.

L2: deeper investigation, incident response, endpoint/network analysis, containment.

L3: advanced threat hunting, detection engineering, malware analysis, complex incidents.

Detection engineer: converts adversary behavior and investigative knowledge into reliable detections with low false-positive rates.

Threat hunter: proactively searches for suspicious behavior without waiting for a high-confidence alert.

3. Networking Fundamentals for SOC

Must-know concepts

OSI/TCP-IP models; IPv4/IPv6; MAC vs IP; ARP; routing; NAT; TCP vs UDP; ports; sockets; stateful firewalls.

DNS: A/AAAA, CNAME, MX, TXT, NS, PTR; recursive resolution; caching; DNS tunneling indicators.

HTTP/S: methods, status codes, headers, cookies, user-agent, redirects, TLS certificates and SNI.

Common ports: 22 SSH, 25 SMTP, 53 DNS, 80 HTTP, 88 Kerberos, 110 POP3, 123 NTP, 135 RPC, 139 NetBIOS, 143 IMAP, 389 LDAP, 443 HTTPS, 445 SMB, 636 LDAPS, 3389 RDP, 5985/5986 WinRM.

Useful network tools

Wireshark / tshark — packet capture and protocol analysis.

tcpdump — command-line packet capture.

Zeek — network security monitoring and rich protocol logs.

Suricata / Snort — IDS/IPS and network signatures.

nmap — authorized network discovery and service enumeration.

Netcat — controlled connectivity testing and troubleshooting.

curl — HTTP/API testing and header inspection.

dig / nslookup — DNS troubleshooting and investigation.

Example commands

```
ipconfig /all
ipconfig /displaydns
nslookup example.com
tracert example.com

ip addr
ip route
ss -tulpn
dig example.com
curl -I https://example.com

sudo tcpdump -i any -nn host 10.0.0.5
sudo tcpdump -i any -nn port 53
```

4. Windows Security for SOC

What to learn

Windows Event Viewer and event channels: Security, System, Application, PowerShell, Windows Defender, Task Scheduler and service-related telemetry.

Authentication: logon/logoff, failed logons, account changes, privilege changes, Kerberos, NTLM and lateral movement concepts.

Processes: parent/child relationships, command lines, executable paths, hashes, signatures, network connections.

Persistence: services, scheduled tasks, startup locations, registry run keys, WMI, DLL/search-order abuse and other ATT&CK; techniques.

Active Directory basics: domains, users, groups, OUs, GPOs, domain controllers, Kerberos, LDAP, privileged groups.

Important Windows/Sysmon telemetry

4624 successful logon; 4625 failed logon; 4688 process creation when process auditing is enabled; 4672 special privileges; 4720 account created; 4728/4732 group membership changes; 4740 account lockout; 4768/4769 Kerberos events; 7045 service installation.

Sysmon can provide process creation, network connection, image load, file creation and other high-value endpoint telemetry depending on configuration.

Useful commands

```
whoami /all
hostname
systeminfo
tasklist /v
netstat -ano
ipconfig /all

Get-WinEvent -LogName Security -MaxEvents 20
Get-Process | Sort-Object CPU -Descending | Select -First 20
Get-NetTCPConnection | Sort-Object RemotePort

Get-FileHash .\suspicious.exe -Algorithm SHA256
Get-Service | Where-Object {$_.Status -eq 'Running'}
```

5. Linux Security for SOC

What to learn

/var/log/auth.log or /var/log/secure for authentication-related activity depending on distribution; journald/systemd; SSH; sudo; cron; services; permissions; users/groups; processes; network sockets.

Understand sudo abuse, SSH brute force, suspicious binaries, unusual child processes, persistence via cron/systemd, privilege escalation indicators and unauthorized accounts.

Useful commands

```
who
w
last
lastb
id
sudo -l
ps aux --sort=-%cpu | head
ss -tuln

journalctl --since '1 hour ago'
journalctl -u ssh
systemctl list-units --type=service --state=running
crontab -l

grep -Ei 'failed|invalid|authentication' /var/log/auth.log | tail -50
find /tmp -type f -mtime -1 -ls
```

6. SIEM — Security Information and Event Management

Major platforms to know

Microsoft Sentinel — cloud SIEM/SOAR ecosystem with KQL.

Splunk Enterprise Security — widely used SIEM/security analytics platform; SPL is central.

IBM QRadar — SIEM with offenses, flows and event analysis.

Elastic Security — SIEM/XDR capabilities on the Elastic Stack; KQL/EQL/Lucene-style searches depending on workflow.

Google Security Operations — Google's security analytics platform, using its own search/detection capabilities.

Other ecosystems: Sumo Logic, LogRhythm, Rapid7 InsightIDR, Exabeam and managed SOC/MDR platforms.

SIEM skills

Log onboarding and parsing; fields; timestamps; normalization; data quality; retention; search; correlation; thresholds; allowlists; suppression; dashboards; alert severity; case management.

Learn one SIEM deeply, then transfer the investigation concepts to others.

Search examples

```
Splunk SPL example:  
index=windows EventCode=4625 | stats count by src_ip, user | sort - count
```

```
Sentinel KQL example:  
SigninLogs  
| where ResultType != 0  
| summarize Failed=count() by IPAddress, UserPrincipalName  
| order by Failed desc
```

```
Elastic/KQL-style example:  
event.category:authentication AND event.outcome:failure
```

7. EDR / XDR / Endpoint Telemetry

Tools and ecosystems

Microsoft Defender for Endpoint — endpoint detection/response, investigation and hunting.

CrowdStrike Falcon — endpoint detection, threat hunting and response.

SentinelOne Singularity — endpoint prevention, detection and response.

VMware Carbon Black — endpoint security and threat hunting.

Elastic Defend — endpoint telemetry and security analytics.

Trend Micro, Sophos, Trellix and other enterprise endpoint security platforms.

What analysts investigate

Process tree; command line; hash; signer; parent process; user; host; persistence; network connections; DNS; file writes; registry activity; prevalence; first/last seen; related alerts.

High-value question: does the process make sense for its parent, user, location, time, and behavior?

8. IDS / IPS / NDR

Tools

Suricata — signature-based and protocol-aware network detection.

Snort — classic IDS/IPS engine and rule ecosystem.

Zeek — metadata-rich network monitoring and scripting.

Corelight — commercial network security platform built around Zeek.

Security appliances from major firewall/network vendors also produce valuable telemetry.

SOC use

Detect scanning, command-and-control patterns, exploitation attempts, suspicious DNS, unusual protocols, data transfer anomalies and lateral movement.

Network alerts are evidence, not automatic proof of compromise. Correlate with endpoint, identity and application telemetry.

9. Threat Intelligence

Concepts

IOC: observable such as IP, domain, URL, hash or email artifact. TTP: adversary behavior or technique. Context determines whether an observable is malicious.

Use multiple sources and reputation context; do not treat one reputation result as definitive.

Tools / sources

VirusTotal — malware/file/URL/domain intelligence and enrichment.

AbuseIPDB — IP reputation and abuse reports.

URLhaus — malicious URL intelligence.

AlienVault OTX — community threat intelligence.

MISP — open-source threat intelligence sharing and management platform.

MITRE ATT&CK; — adversary tactics, techniques and procedures.

STIX/TAXII — structured threat intelligence representation and exchange.

Enrichment workflow

Extract IOC → normalize → check reputation/context → search SIEM/EDR for internal occurrences → pivot on related infrastructure/users/processes → document confidence and evidence.

10. MITRE ATT&CK; for SOC Analysts

Tactics to recognize

Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, Impact.

Why it matters

ATT&CK; provides a common language for detections, threat hunting, incident reports and coverage mapping.

Map observed behavior to techniques rather than relying only on malware names.

High-value techniques to study

Phishing; Valid Accounts; Command and Scripting Interpreter; PowerShell; Windows Management Instrumentation; Scheduled Task/Job; Create or Modify System Process; OS Credential Dumping; Remote Services; SMB/Windows Admin Shares; RDP; Data Staged; Exfiltration Over Web Service; C2 over common protocols.

11. Email Security & Phishing Analysis

Tools

Microsoft Defender for Office 365 / email security portals; Proofpoint; Mimecast; secure email gateways; sandboxing platforms.

Message header analysis tools; URL/domain reputation services; sandboxing; SIEM/EDR correlation.

Analyze

Sender/from vs return-path; authentication results; SPF/DKIM/DMARC; Received headers; sending IP; reply-to; URLs; attachments; file hashes; lookalike domains; user interaction; mailbox search for other recipients.

Phishing triage checklist

Confirm whether the message reached multiple users; identify clicks; isolate malicious messages if authorized; search for matching IOCs; check endpoint activity; reset credentials when compromise is supported by evidence.

12. Cloud Security for SOC

AWS

CloudTrail; VPC Flow Logs; GuardDuty; IAM; S3 access logging; CloudWatch; Security Hub; EC2 telemetry.

Microsoft Azure

Microsoft Entra ID; Azure Activity Logs; Microsoft Defender for Cloud; Microsoft Sentinel; NSG flow logs; Key Vault; Storage logs.

Google Cloud

Cloud Audit Logs; Security Command Center; VPC Flow Logs; IAM; Cloud Logging.

Cloud SOC skills

Identity-first investigation; impossible travel/anomalous sign-ins; privilege changes; access-key activity; suspicious API calls; public exposure; unusual resource creation; data access and exfiltration indicators.

13. Vulnerability Management

Tools

Tenable Nessus / Tenable Vulnerability Management; Qualys VM; Rapid7 InsightVM; OpenVAS/Greenbone; cloud-native vulnerability scanners.

SOC relevance

Vulnerability data helps prioritize alerts and incidents. A critical alert on an unpatched internet-facing asset deserves different urgency from the same behavior on an isolated lab host.

Concepts

CVE, CVSS, CWE, exploitability, exposure, asset criticality, compensating controls, patch verification, remediation SLA.

14. Incident Response

Lifecycle

Preparation → Detection/Analysis → Containment → Eradication → Recovery → Lessons Learned.

Maintain evidence integrity and follow organizational procedures. Preserve timestamps, relevant logs, hashes, screenshots and case notes where policy requires.

Containment examples

Disable or reset compromised accounts; isolate endpoints; block malicious indicators; revoke tokens/keys; quarantine email; block malicious domains/IPs; terminate malicious sessions.

Do not take destructive actions without authorization and evidence unless an established emergency procedure allows it.

Incident report structure

Executive summary; timeline; affected assets/users; initial access; attacker behavior; IOCs; ATT&CK; mapping; evidence; containment; eradication; recovery; root cause; impact; recommendations.

15. Detection Engineering

Detection lifecycle

Hypothesis → data source → observable behavior → query/rule → test → tune → deploy → monitor false positives → measure coverage → maintain.

Prefer behavior-based detections over brittle indicators when feasible.

Detection quality

High signal, explainable, actionable, documented, tested and resilient to normal environmental changes.

Track false positives, alert volume, mean time to triage, detection coverage and data-source health.

Sigma

Sigma is a generic rule format for describing SIEM detections and converting them to platform-specific queries.

Learn fields, logsource, detection logic, conditions, false positives and level.

16. SOAR & Automation

Tools / concepts

Microsoft Sentinel automation; Splunk SOAR; Cortex XSOAR; Tines; Shuffle; Torq and other workflow automation platforms.

Common integrations: SIEM, EDR, ticketing, email, threat intel, identity provider, firewall and cloud APIs.

Automation examples

Enrich an IP/domain/hash; open a case; add context; notify analyst; quarantine email; isolate endpoint; disable account — only after appropriate approval and confidence.

Use automation to reduce repetitive work, not to blindly automate high-impact decisions.

17. Malware Analysis Basics

Tools

VirusTotal; Any.Run; CAPE/Cuckoo sandbox ecosystems; Ghidra; strings; YARA; Detect It Easy; PE-bear; Wireshark; Procmon; Autoruns; TCPView; Sysmon.

Static analysis

Hash; file type; metadata; strings; imports; signatures; embedded URLs/domains; suspicious sections; entropy; packer indicators.

Dynamic analysis

Observe process tree, file changes, registry changes, network connections, DNS, persistence and behavior in an isolated lab.

Safety

Use isolated virtual machines and snapshots. Never execute unknown malware on a production machine or personal network.

18. Forensics & Investigation Tools

Windows

Sysinternals: Autoruns, Process Explorer, Procmon, TCPView, Sigcheck, PsList, Strings.

Windows Event Viewer, Windows Terminal/PowerShell, KAPE and other forensic collection frameworks.

Disk/memory

Autopsy/The Sleuth Kit; Volatility/Volatility 3; FTK Imager and enterprise forensic platforms.

Investigation principle

Build a timeline and correlate independent evidence sources. Avoid concluding compromise from a single suspicious artifact.

19. Useful Security Tool Stack

Recommended beginner lab stack

VirtualBox/VMware/Hyper-V; Windows evaluation VM; Ubuntu/Kali VM; Wazuh; Sysmon; Elastic or Splunk Free/Developer options where applicable; Wireshark; Zeek; Suricata; Sigma; YARA; Python; PowerShell; Git.

Tool matrix

Category	Examples	Learn first
SIEM	Sentinel, Splunk, Elastic, QRadar	1 SIEM deeply
EDR/XDR	Defender, CrowdStrike, SentinelOne	1 platform concepts
NDR/IDS	Zeek, Suricata, Snort	Zeek + Suricata
Packet analysis	Wireshark, tshark, tcpdump	Wireshark
Threat intel	VirusTotal, OTX, MISP, AbuseIPDB	VirusTotal + ATT&CK;
Detection	Sigma, YARA	Sigma
IR	Sysinternals, KAPE, Velociraptor	Sysinternals
Forensics	Volatility, Autopsy	Volatility
Vulnerability	Nessus, Qualys, Greenbone	Nessus/Greenbone
Automation	Python, PowerShell, SOAR	Python + PowerShell
Cloud	Sentinel/Defender, GuardDuty, Security Hub	One cloud

20. Programming & Scripting for SOC

Python

Learn variables, functions, files, JSON/CSV, regex, requests, argparse, logging, error handling, APIs and basic data analysis.

Useful projects: IOC extractor, hash checker, log parser, IP enrichment script, suspicious URL analyzer, alert deduplicator.

PowerShell

Learn Get-WinEvent, Get-Process, Get-Service, Get-NetTCPConnection, CIM/WMI, objects, filtering, exporting CSV/JSON and REST APIs.

Bash

Learn grep, awk, sed, cut, sort, uniq, find, xargs, curl, jq, pipes/redirection and shell scripting.

Example IOC extraction

```
import re
text = open('sample.log', errors='ignore').read()
ips = sorted(set(re.findall(r'\b(?:\d{1,3}\.){3}\d{1,3}\b', text)))
for ip in ips:
    print(ip)
```

21. Log Analysis

What to extract

Timestamp, hostname, username, source/destination IP, port, process, parent process, command line, event ID, action, status, file hash, URL/domain, user agent, authentication method.

Investigation pivots

IP → users → hosts → events; user → logons → processes → network; hash → hosts → process trees → network; domain → DNS → proxy → endpoint; alert → surrounding time window → related alerts.

Timeline method

Start with the first known suspicious event, expand backward to identify initial access and forward to determine impact and persistence.

22. SOC Alert Triage Playbook

10-step triage

1. Read the alert title and rule logic.
2. Identify asset/user and business criticality.
3. Validate timestamp and timezone.
4. Identify source and destination.
5. Enrich IOCs.
6. Check endpoint process tree or relevant log context.
7. Search for related activity before/after the alert.
8. Determine benign, suspicious, or malicious with confidence.
9. Contain/escalate according to playbook.
10. Document evidence and next action.

Severity thinking

Consider confidence × impact × scope × exposure. A suspicious event on a domain controller or privileged account may require higher priority than the same event on a test machine.

23. Common SOC Use Cases

Authentication

Brute force; password spraying; impossible travel; MFA anomalies; new device; suspicious OAuth consent; privileged login; disabled-account use.

Endpoint

Malware execution; LOLBins; PowerShell abuse; suspicious parent-child process; persistence; credential dumping; ransomware behavior; unsigned binaries.

Network

Port scanning; beaconing; DNS anomalies; suspicious outbound traffic; C2; lateral movement; unusual data transfer.

Identity/Cloud

New privileged role; access-key creation; suspicious API calls; public storage exposure; unusual resource creation; mass downloads.

Email

Phishing; malicious attachment; credential-harvesting URL; spoofing; business email compromise; mailbox rule abuse.

24. Common False Positives

Examples

Administrative scripts; software deployment; vulnerability scanners; backup systems; monitoring tools; legitimate remote administration; security testing; developer tooling; scheduled tasks.

Do not simply suppress an alert. Document the reason, scope the exception narrowly, set an expiration where possible, and continue monitoring.

25. SOC Documentation & Ticketing

Good ticket format

Title; detection; time; affected host/user; severity; evidence; investigation steps; IOC list; impact; actions taken; escalation; current status; recommended next action.

Writing style

Use precise timestamps, avoid unsupported assumptions, distinguish observed facts from hypotheses, and include enough evidence that another analyst can reproduce the reasoning.

26. Home SOC Lab

Suggested topology

Host machine → hypervisor → Windows VM + Linux VM + security monitoring VM. Keep the lab isolated from production networks.

Generate benign test activity: failed logins, PowerShell commands, DNS queries, web requests, file creation, process execution, SSH login attempts and network connections.

Lab projects

1. Install Sysmon and collect Windows telemetry.
2. Send logs to Wazuh/Elastic/Splunk/Sentinel lab.
3. Create a failed-login detection.
4. Investigate a suspicious PowerShell process.
5. Capture traffic with Wireshark.
6. Create Suricata/Zeek telemetry.
7. Write a Sigma rule.

8. Enrich an IOC with threat-intel sources.
9. Build a phishing investigation case.
10. Write a complete incident report.

27. 12-Week SOC Analyst Roadmap

Weeks 1-2: Networking

TCP/IP, DNS, HTTP/S, ports, firewalls, packet analysis, Wireshark and basic Linux networking.

Weeks 3-4: Windows/Linux

Windows logs, Sysmon, PowerShell, AD basics; Linux auth logs, SSH, processes, permissions and shell.

Weeks 5-6: SIEM

Pick Sentinel, Splunk or Elastic. Learn ingestion, searches, fields, dashboards, alerts and investigations.

Weeks 7-8: EDR + Threat Intel

Learn process trees, endpoint investigation, IOC enrichment, ATT&CK; and common attack chains.

Weeks 9-10: Incident Response + Detection

Triage, containment, reporting, Sigma, detection tuning and basic threat hunting.

Weeks 11-12: Cloud + Automation + Portfolio

Choose AWS/Azure/GCP; learn identity/audit logs; build Python/PowerShell automation; publish sanitized case studies and detection rules.

Daily schedule

60-90 min theory + 90 min hands-on lab + 30 min investigation/write-up. Consistency matters more than collecting certificates.

28. Certifications & Learning Direction

Entry / foundation

CompTIA Security+ is a common foundation option. Google Cybersecurity Certificate can help absolute beginners. Vendor-specific training can complement hands-on skills.

SOC-focused

Microsoft SC-200 is relevant to Microsoft security operations. Splunk certifications/training are useful in Splunk-heavy environments. Blue-team practical certifications can be useful when paired with real lab work.

Important

Certifications do not replace investigation ability. Build a lab portfolio with detections, incident reports, queries and screenshots/notes that do not expose real sensitive data.

29. Interview Preparation

Questions you should be able to answer

What is a SIEM? What is EDR? Difference between IDS and IPS?

Explain TCP handshake. What happens during DNS resolution?

How would you investigate a brute-force alert?

What is the difference between an IOC and a TTP?

How do you investigate suspicious PowerShell?

What Windows events are useful for authentication and process creation?

How would you determine whether an alert is a false positive?

What is lateral movement? Give examples.

How do you respond to a compromised endpoint?

What is MITRE ATT&CK; and how do you use it?

How would you investigate a phishing email?

What logs would you need to investigate suspicious cloud sign-in activity?

Interview answer framework

State what you observed → identify the evidence needed → explain investigation pivots → state containment/escalation → explain documentation and lessons learned.

30. SOC Career & Job-Ready Checklist

You are job-ready when you can

Read and explain common Windows/Linux/network logs.

Perform a structured SIEM investigation.

Use an EDR process tree and endpoint timeline.

Analyze DNS/HTTP/packet evidence.

Map behavior to ATT&CK.;

Enrich and pivot on IOCs.

Write a clear incident ticket.

Explain containment and escalation decisions.

Write basic Sigma/YARA rules.

Use Python/PowerShell/Bash for repetitive tasks.

Demonstrate at least 5–10 documented lab investigations.

Portfolio projects

Project 1: Failed-login/password-spray investigation.

Project 2: Suspicious PowerShell investigation.

Project 3: Phishing email investigation.

Project 4: Malware/endpoint timeline investigation in an isolated lab.

Project 5: Network scan/IDS investigation.

Project 6: Cloud identity anomaly investigation.

Project 7: Sigma detection pack with test logs.

Project 8: Python IOC enrichment tool.

31. Quick Reference — SOC Analyst Tool Map

Need	First tools to reach for
Packet capture	Wireshark / tcpdump
DNS investigation	dig / nslookup / Wireshark / Zeek
Windows process investigation	Sysmon / Process Explorer / EDR
Windows logs	Event Viewer / PowerShell / SIEM
Linux auth	journalctl / auth.log / secure
SIEM investigation	Sentinel / Splunk / Elastic / QRadar
Endpoint investigation	Defender / CrowdStrike / SentinelOne
Threat intel	VirusTotal / OTX / MISP / ATT&CK;
Network detection	Zeek / Suricata / Snort
File analysis	hash tools / strings / sandbox / Ghidra
Memory forensics	Volatility
Disk forensics	Autopsy / The Sleuth Kit
Detection rules	Sigma / YARA
Automation	Python / PowerShell / Bash / SOAR
Vulnerability	Nessus / Qualys / Greenbone
Cloud investigation	CloudTrail / Entra / Azure Activity Logs / Cloud Audit Logs

32. Final SOC Analyst Checklist

Daily practice

Read one incident write-up; investigate one lab alert; write one query; learn one ATT&CK; technique; practice one command; document what you found.

Avoid these mistakes

Tool collecting without understanding; copying queries without understanding fields; treating reputation services as absolute truth; closing alerts without evidence; overusing broad allowlists; ignoring asset criticality; failing to document timestamps and pivots.

Core principle

Think like an investigator: What happened? How do I know? What else should I expect to see? What is the impact? What should happen next?

Appendix A — Essential SOC Glossary

Term	Meaning
SIEM	Central security log collection, search, correlation and alerting platform.
EDR	Endpoint detection and response platform focused on endpoint telemetry and response.
XDR	Cross-domain detection/response combining endpoint and other security telemetry.
IOC	Indicator of compromise such as a hash, IP, domain or URL.
TTP	Tactics, techniques and procedures describing adversary behavior.
SOAR	Security orchestration, automation and response.
IDS/IPS	Intrusion detection/prevention systems for network activity.
NDR	Network detection and response.
IOC pivot	Using one artifact to find related activity across telemetry.
False positive	Alert that matches a rule but represents benign activity.
True positive	Alert correctly identifying suspicious/malicious activity.
MTTD	Mean time to detect.
MTTR	Mean time to respond/recover, depending on organizational definition.
CVE	Public identifier for a vulnerability.
CVSS	Common Vulnerability Scoring System.
YARA	Rule language/tooling for identifying patterns in files or memory.
Sigma	Generic detection-rule format for log/SIEM detections.

Appendix B — Suggested Study Order

Networking → Windows/Linux → Logs → SIEM → EDR → Threat Intel → ATT&CK; → Incident Response → Detection Engineering → Cloud → Automation → Advanced hunting/forensics.

Recommended learning principle: understand the telemetry first, then learn how each product collects and queries it.

End of Handbook